

Publication Retractions Among Biomedical Research Publications Funded and Not Funded by the National Institutes of Health (NIH)

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Abstract

I present descriptive analyses of retractions of biomedical research publications funded and not funded by the National Institutes of Health (NIH). I identified 24.4 million papers published from 1981 to 2024, 2.6 million funded at least in part by the NIH. There were 21,319 retractions (0.09%). Retraction rates were higher for non-NIH funded papers (0.09% vs 0.06%) and a greater proportion of non-NIH retractions were linked to some form of investigator misconduct (86% vs 80%). NIH-funded papers were more likely to be retracted because of typical types of misconduct, like institutional findings, image duplication, and falsification or fabrication of data. Non-NIH funded papers were more likely to be retracted because of third-party concerns, such as journal-instigated investigations and paper mills, which were rarely seen with NIH-funded papers. Consistent with previous literature, most non-NIH funded retracted papers came from China. NIH-funded papers saw a much longer lagtime between publication and retraction (median times 2008 days versus 695 days for non-NIH funded papers). Secular trends differed as retraction rates for non-NIH funded papers increased in the 2010s and early 2020s, largely due to paper mill activities. Retraction rates for NIH-funded papers have declined since the early 2010s, but these findings should be interpreted with caution given the long lagtimes to retraction.

Background

Publication retractions offer a window into scientific misconduct as most retractions are due to some kind of investigator misbehavior.^(1–3) To my knowledge, there has not been a recent assessment of retractions according to source of funding. Stern et al reported in 2012 that NIH-funded papers retracted for misconduct accounted for less than one percent of the NIH budget.⁽⁴⁾

The Retraction Watch database, launched in 2018, provides accurate and detailed information on academic retractions, including reasons for retraction, time from publication to retraction, and country (countries) of origin.⁽⁵⁾ Investigators have used the database to conduct research on retractions and scientific misconduct.^(2, 3, 6) I leveraged the Retraction Watch database to describe reasons and trends in research article retractions according to source of funding.

Data and Methods

The NIH Office of Portfolio Analysis (or OPA) iCite product regularly releases detailed data on all papers in Pubmed published after 1980.⁽⁷⁾ OPA data includes the relative citation ratio (RCR) metric, which measures citation impact after accounting for field and year of publication.⁽⁸⁾ OPA also categorizes papers by whether they have human, animal, and/or molecular/cellular MeSH terms.⁽⁹⁾ Over 95% of NIH-funded papers have at least one such term. I downloaded the OPA database on June 11, 2026.

The NIH Office of Extramural Research maintains NIH’s ExPORTER website (see <https://reporter.nih.gov/exporter>), which includes yearly files that link publications to NIH awards.

Using the OPA database, I identified papers published between 1981 and 2024 that had at least one human, animal, or molecular/cellular MeSH term. As OPA has done, I classified papers with no human MeSH terms as describing fundamental science and those with only human MeSH terms as describing human-focused science.⁽⁹⁾ Mixed science papers were those with human MeSH terms and at least one animal and/or molecular/cellular term. I merged the OPA database with data from NIH ExPORTER to identify which papers were at least partially funded by NIH. I took steps to assure correct classification of NIH-funded papers, since ExPORTER includes data on papers and awards that come from some non-NIH agencies (e.g., the Agency for Health Research and Quality and the US Department of Veterans Affairs).

Using the Retraction Watch database (as downloaded on April 10, 2026), I identified papers that underwent retraction (i.e., not just expression of concern). I merged the resulting table with the OPA table and kept those papers which either OPA or Retraction Watch called a research article.

I designated a retraction as due to misconduct if at least one of the reasons for retraction included fabrication, falsification, plagiarism; breaches of policy on the part of an author; concerns about data or images; violations related to peer review, animal welfare, human subject protections, or informed consent; or other reasons related to misbehavior, such as paper mills, forged authorship, or a hoax paper.

I used Julius AI ⁽¹⁰⁾ for help in writing R-markdown code ⁽¹¹⁾, but I did not use AI for writing manuscript text.

Results

After inclusions and exclusions, my working data frame included 24,375,777 research papers, of which 21,319 were retracted (0.09%). Among the 2,587,674 NIH-funded papers, 1,658 (0.06%) were retracted; 1,333 (80.40%) were retracted due to some form of misconduct. Among the 21,788,103 non-NIH-funded papers, 19,661 (0.09%) were retracted; 16,907 (85.99%) were retracted due to some form misconduct.

Comparison of Retracted and Non-Retracted Papers

Retracted papers were less likely to be linked to NIH funding and were less likely to involve human-focused science or clinical studies (Table 1). They were less translational (that is, their Approximate Potential to Translate metric⁽⁹⁾ was lower) and had lower citation impact as assessed by mean relative citation ratio or by having a relative citation ratio above the 95th percentile.

Reasons for Retraction by NIH Funding Status

Tables 2 and 3 show the most common reasons for retraction among retracted papers, separately for NIH-funded papers and non-NIH funded papers. It is important to note that the Retraction Watch database reasons are not mutually exclusive, so a single paper typically has multiple reasons for retraction. NIH-funded papers were more likely retracted due to typical scientific misconduct, like institutional findings, image duplication, and falsification or fabrication of data. Non-NIH-funded papers were more likely retracted because of third-party concerns, like journal-instigated investigations, peer review violations, and paper mills. Among NIH-funded retracted papers, only 4 (0.24%) had at least one listed retraction reason of “Paper Mill.”

There were also marked differences in time to retraction (Tables 2 and 3). Irrespective of reason, NIH-funded papers took much longer to retract. Among all retracted papers with available publication and retraction dates, the median time to retraction was 2,008 days (IQR 856-3,518) for NIH-funded papers and 695 days (IQR 385-1,539) for non-NIH-funded papers.

Among papers retracted for non-misconduct reasons, common reasons for retraction included errors or unreliability. Time to retraction was relatively short for both NIH-funded and non-NIH funded papers (Tables 2a and 3a).

Retraction Rates over Time

For papers published before the year 2000, retraction was a rare event irrespective of funding source (Figure 1). For papers published in the 2000s and 2010s, retraction rates increased, somewhat more among NIH-funded papers. For papers published since 2015, retraction rates increased dramatically for non-NIH funded papers, especially those published in 2022, likely due to exposures of paper mills and peer review fraud schemes. Retraction rates for NIH-funded papers declined, but interpretation of this observation requires caution given the long lag periods between publication and retraction for NIH-funded papers (Table 2).

The trends were similar, though not as marked, even after excluding papers that were retracted because of paper mills (Figure 2).

Funding and Country of Origin for Retracted Papers.

As expected, nearly all NIH-funded retracted papers originated in the United States, though nearly 9% had at least one link to China (Table 4). Among non-NIH funded papers, over sixty percent originated in China (Table 5), an observation consistent with prior literature.⁽¹²⁾ Note that countries are not mutually exclusive, so one paper may be linked to multiple countries.

The reasons for retractions among American and Chinese papers largely reflected the differences in reasons for NIH-funded and non-NIH funded papers (Tables 6 and 7).

Discussion

Retraction patterns differ for NIH-funded and non-NIH funded papers. Retracted NIH-funded papers feature typical forms of scientific misconduct, whereas non-NIH funded retracted papers, which come mainly from China, feature third-party issues, including paper mills and peer review violations. The majority of retractions were due to some form of misconduct, but misconduct was more common among non-NIH funded retracted papers.

NIH-funded papers take much longer to retract, with the difference in retraction times between NIH- and non-NIH funded papers measured in *years*. The time differential likely reflects the different reasons for retraction, as scientific misconduct investigations and implementation of remedies typically take many years. Another source of delay stems from complex journal editorial processes as well as reluctance among some editors to draw attention to problematic papers.⁽¹³⁾

Over the past 10 to 15 years, there has been increasing concern about contamination of the scientific literature from papers produced by paper mills⁽¹⁴⁾, defined by Parker et al as “fraudulent organizations that make money by writing fake manuscripts and offering authorship slots for sale to academic customers.”⁽¹⁵⁾ The current analyses suggest that NIH-funded investigators have not materially used paper mills, at least not for papers that they link to NIH grants. Paper mills have emerged as a concerning source of fraudulent papers coming from China, which has increasingly recognized its research integrity problems.^(16, 17)

My analyses extend on previous analyses of NIH-linked retractions. Stern et al focused on investigators censured by the US Department of Health and Human Services Office of Research Integrity (ORI) and found that affected investigators saw marked declines in research funding and publication productivity.⁽⁴⁾ Peterson et al focused on retraction notices for 137 NIH-funded articles published between 2019 and 2023. They found that the most common reasons for retraction were data integrity and problems with images and figures.⁽¹⁸⁾ One of the key strengths of the Retraction Watch database is its careful characterization of retraction reasons, reasons that go beyond what is stated in retraction notices.⁽¹⁴⁾

An important limitation of these analyses is the absence of data on detailed timelines. Retractions often involve messy, complex, and expensive series of events, ranging from nascent allegations through inquiries, investigations, appeals, litigation, and extensive communications between affected parties. I characterized funding as NIH or not NIH, but did not analyze outcomes according to other large funders, like the National Natural Science Foundation of China. Another limitation is that retractions are extreme events and may only

reflect a fraction of misconduct. Surveys of scientists and careful assessments of published papers indicate that the true prevalence of misconduct substantially exceeds retraction rates.(19, 20)

Despite these limitations, the current analyses offer insights into the nature of retractions according to source of funding and country of origin. Because of the long time lags between publication and retraction for NIH-funded studies, seeming declines in retraction rates should be interpreted with caution. We are fortunate to have robust data on biomedical publications and on retractions, but more routinely collected data would help policy makers better understand real-time trends in research integrity and better address integrity breaches in a timely manner.(21)

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Data and Code

All data and code are publicly available. OPA data is at <https://support.icite.nih.gov/hc/en-us/articles/9513563045787-Bulk-Data-and-API>. NIH ExPORTER data is at <https://reporter.nih.gov/exporter>. The Retraction Watch database is at <https://retractiondatabase.org/RetractionSearch.aspx?>. I posted R-markdown replication code at https://github.com/lauermichaels/Retraction_project.

Table 1: Comparison of Retracted and Non-Retracted Papers

Highly-cited is defined as non-missing relative citation ratio at or above full-database 95th percentile: 3.70

Variable	Retraction status	
	Retracted	Non-retracted
N papers	21,319	24,354,458
NIH funding	7.78% (1,658/21,319)	10.62% (2,586,016/24,354,458)
Type of science - Fundamental	29.26% (6,237/21,319)	29.75% (7,245,769/24,354,458)
Type of science - Mixed	40.19% (8,569/21,319)	21.18% (5,157,505/24,354,458)
Type of science - Human-focused	30.55% (6,513/21,319)	49.07% (11,951,184/24,354,458)
Clinical Study	2.34% (498/21,319)	4.66% (1,134,557/24,354,458)
Author count, mean (SD)	5.76 (3.48)	5.45 (5.67)
Approximate Potential to Translate, mean (SD)	0.17 (0.19)	0.24 (0.26)
Number of citations, mean (SD)	22.08 (66.65)	25.94 (103.95)
Relative citation ratio, mean (SD)	0.93 (2.36)	1.10 (4.24)
Highly-cited (RCR $\geq$ 95th percentile, 3.70)	3.08% (655/21,287)	5.00% (1,217,259/24,327,238)

Categorical variables are shown as percent (n/N). Continuous variables are shown as mean (SD). The denominator for highly-cited includes only papers with non-missing relative citation ratio.

Table 2: Most common reasons for and time to retraction among NIH-funded papers

Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency		Median days (IQR)
		Papers	% retracted	
1	Investigation by Company/Institution	605	36.5	2,533 (1,538-3,833)
2	Duplication of/in Image	545	32.9	3,149 (1,994-4,580)
3	Misconduct - Official Investigation(s) and/or Finding(s)	358	21.6	2,173 (1,234-3,466)
4	Falsification/Fabrication of Data	348	21.0	2,070 (1,210-3,512)
5	Misconduct by Author	340	20.5	2,008 (1,156-3,387)
6	Concerns/Issues about Data	323	19.5	2,325 (968-3,500)
7	Unreliable Results and/or Conclusions	294	17.7	2,026 (849-4,135)
8	Investigation by Journal/Publisher	252	15.2	3,062 (1,582-4,881)
9	Manipulation of Images	245	14.8	2,898 (1,972-4,426)
10	Concerns/Issues about Image	222	13.4	3,175 (1,468-4,630)
11	Falsification/Fabrication of Image	212	12.8	2,319 (1,328-3,750)
12	Investigation by ORI	191	11.5	1,877 (894-3,215)
13	Unreliable Data	175	10.6	2,435 (1,204-4,094)
14	Original Data and/or Images not Provided and/or not Available	161	9.7	3,790 (2,382-5,150)
15	Error in Data	143	8.6	1,034 (478-1,864)

% retracted uses NIH-funded retracted papers as the denominator. Time to retraction is shown as median days (IQR).

Table 3: Most common reasons for and time to retraction among non-NIH-funded papers

Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency		Median days (IQR)
		Papers	% retracted	
1	Investigation by Journal/Publisher	9,353	47.6	603 (414-1,187)
2	Unreliable Results and/or Conclusions	8,310	42.3	587 (414-1,009)
3	Investigation by Third Party	7,444	37.9	620 (434-1,233)
4	Concerns/Issues about Data	7,008	35.6	600 (394-1,203)
5	Paper Mill	4,837	24.6	617 (462-1,014)
6	Concerns/Issues about Peer Review	4,278	21.8	466 (361-615)
7	Concerns/Issues about Referencing/Attributions	4,192	21.3	480 (369-636)
8	Duplication of/in Image	3,772	19.2	1,604 (974-2,562)
9	Concerns/Issues about Results and/or Conclusions	3,522	17.9	531 (393-770)
10	Computer-Aided Content or Computer-Generated Content	2,468	12.6	510 (397-655)
11	Concerns/Issues about Image	2,129	10.8	1,548 (857-2,634)
12	Compromised Peer Review	2,037	10.4	569 (441-773)
13	Original Data and/or Images not Provided and/or not Available	1,775	9.0	1,358 (779-2,440)
14	Lack of IRB/IACUC Approval and/or Compliance	1,575	8.0	504 (375-740)
15	Author Unresponsive	1,493	7.6	1,548 (830-2,362)

% retracted uses non-NIH-funded retracted papers as the denominator. Time to retraction is shown as median days (IQR).

Table 2a: Most common reasons for and time to retraction among NIH-funded papers not retracted due to misconduct

Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency		Median days (IQR)
		Papers	% retracted	
1	Error in Results and/or Conclusions	86	26.5	660 (370-1,514)
2	Error in Data	83	25.5	770 (394-1,486)
3	Unreliable Results and/or Conclusions	72	22.2	824 (343-1,438)
4	Results Not Reproducible	71	21.8	836 (490-1,490)
5	Error in Analyses	68	20.9	637 (326-1,470)
6	Error in Methods	64	19.7	836 (378-1,292)
7	Error in Image	39	12.0	1,037 (653-1,988)
8	Unreliable Data	30	9.2	832 (441-1,475)
9	Notice - Limited or No Information	29	8.9	329 (112-1,269)
10	Retract and Replace	19	5.8	462 (268-897)
11	Error in Cell Lines/Tissues	15	4.6	637 (366-1,564)
12	Contamination of Cell Lines/Tissues	12	3.7	546 (418-1,058)
13	Error by Journal/Publisher	12	3.7	30 (0-30)
14	Unreliable Image	12	3.7	1,049 (632-1,966)
15	Upgrade/Update of Prior Notice(s)	12	3.7	2,293 (978-2,562)

% retracted uses NIH-funded retracted papers not classified as misconduct retractions as the denominator. Time to retraction is shown as median days (IQR).

Table 3a: Most common reasons for and time to retraction among non-NIH-funded papers not retracted due to misconduct

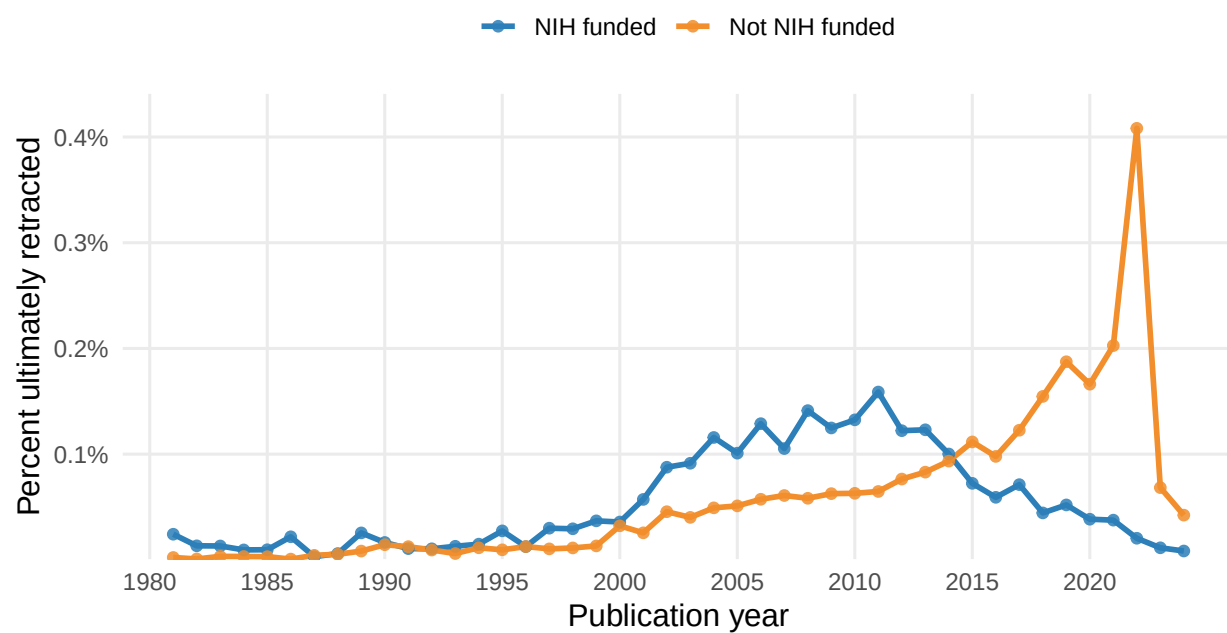
Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency		Median days (IQR)
		Papers	% retracted	
1	Unreliable Results and/or Conclusions	535	19.4	604 (242-1,173)
2	Error in Data	510	18.5	488 (194-1,035)
3	Error in Analyses	450	16.3	455 (200-954)
4	Notice - Limited or No Information	448	16.3	214 (31-768)
5	Error in Results and/or Conclusions	443	16.1	511 (204-998)
6	Error in Methods	366	13.3	419 (179-968)
7	Results Not Reproducible	309	11.2	693 (311-1,160)
8	Removed	306	11.1	231 (1-1,690)
9	Retract and Replace	266	9.7	2,098 (408-3,719)
10	Withdrawn as Out of Date	226	8.2	2,756 (1,901-4,028)
11	Date of Article and/or Notice Unknown	183	6.6	0 (0-174)
12	Unreliable Data	179	6.5	557 (210-1,184)
13	Error in Image	157	5.7	773 (305-1,512)
14	Error by Journal/Publisher	128	4.6	52 (21-181)
15	Concerns/Issues about Authorship/Affiliation	119	4.3	399 (186-666)

% retracted uses non-NIH-funded retracted papers not classified as misconduct retractions as the denominator. Time to retraction is shown as median days (IQR).

Figure 1: Percent of papers retracted by publication year

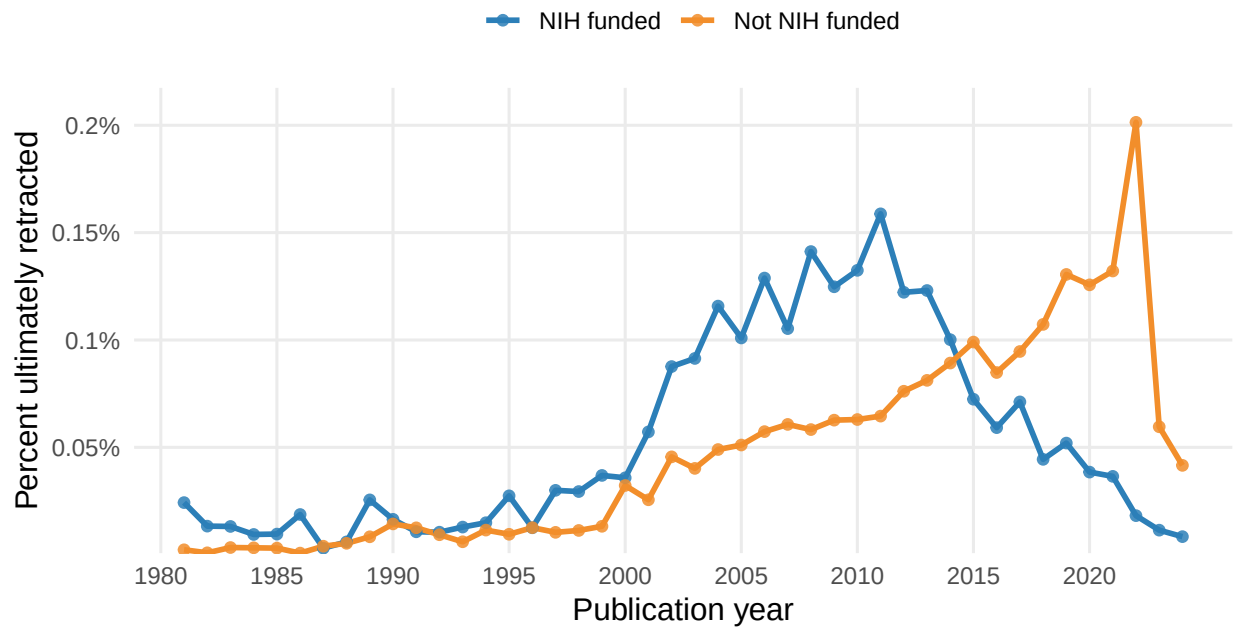
Retraction percentage calculated within publication year and NIH funding group



Note: Recent publication years have less follow-up time for retractions to occur.

Figure 2: Percent of papers retracted by publication year

Excluding papers retracted for paper mills



Note: Recent publication years have less follow-up time for retractions to occur.

Table 4: Most common countries among NIH-funded retracted papers*Countries are not mutually exclusive; one paper can be linked to multiple countries.*

Rank	Country	Frequency	
		Papers	% retracted
1	United States	1,607	96.9
2	China	142	8.6
3	Japan	55	3.3
4	Italy	49	3.0
5	United Kingdom	42	2.5
6	Germany	34	2.1
7	South Korea	26	1.6
8	France	24	1.4
9	Canada	23	1.4
10	India	20	1.2
11	Switzerland	17	1.0
12	Australia	15	0.9
13	Taiwan	14	0.8
14	Greece	13	0.8
15	Spain	13	0.8

% retracted uses NIH-funded retracted papers as the denominator.

Table 5: Most common countries among non-NIH-funded retracted papers*Countries are not mutually exclusive; one paper can be linked to multiple countries.*

Rank	Country	Frequency	
		Papers	% retracted
1	China	11,922	60.6
2	United States	1,792	9.1
3	India	1,279	6.5
4	Saudi Arabia	820	4.2
5	United Kingdom	748	3.8
6	Japan	682	3.5
7	Pakistan	543	2.8
8	Germany	526	2.7
9	Iran	482	2.5
10	South Korea	476	2.4
11	Italy	419	2.1
12	Egypt	393	2.0
13	Canada	347	1.8
14	Australia	306	1.6
15	France	297	1.5

% retracted uses non-NIH-funded retracted papers as the denominator.

Table 6: Most common reasons for retraction among United States-associated papers

Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency	
		Papers	% retracted
1	Investigation by Company/Institution	876	25.8
2	Duplication of/in Image	855	25.2
3	Unreliable Results and/or Conclusions	738	21.7
4	Concerns/Issues about Data	660	19.4
5	Investigation by Journal/Publisher	646	19.0
6	Misconduct - Official Investigation(s) and/or Finding(s)	487	14.3
7	Misconduct by Author	477	14.0
8	Falsification/Fabrication of Data	454	13.4
9	Concerns/Issues about Image	381	11.2
10	Unreliable Data	362	10.7
11	Manipulation of Images	361	10.6
12	Investigation by Third Party	310	9.1
13	Error in Data	305	9.0
14	Falsification/Fabrication of Image	266	7.8
15	Objections by Author(s)	261	7.7

% retracted uses United States-associated retracted papers as the denominator.

Table 7: Most common reasons for retraction among China-associated papers

Reasons are not mutually exclusive; one paper can have multiple reasons.

Rank	Retraction reason	Frequency	
		Papers	% retracted
1	Investigation by Journal/Publisher	7,162	59.4
2	Unreliable Results and/or Conclusions	6,294	52.2
3	Investigation by Third Party	6,284	52.1
4	Concerns/Issues about Data	5,512	45.7
5	Paper Mill	4,402	36.5
6	Concerns/Issues about Peer Review	3,766	31.2
7	Concerns/Issues about Referencing/Attributions	3,589	29.7
8	Concerns/Issues about Results and/or Conclusions	2,809	23.3
9	Duplication of/in Image	2,525	20.9
10	Computer-Aided Content or Computer-Generated Content	2,165	17.9
11	Compromised Peer Review	1,466	12.2
12	Concerns/Issues about Image	1,392	11.5
13	Original Data and/or Images not Provided and/or not Available	1,284	10.6
14	Author Unresponsive	1,279	10.6
15	Lack of IRB/IACUC Approval and/or Compliance	1,257	10.4

% retracted uses China-associated retracted papers as the denominator.